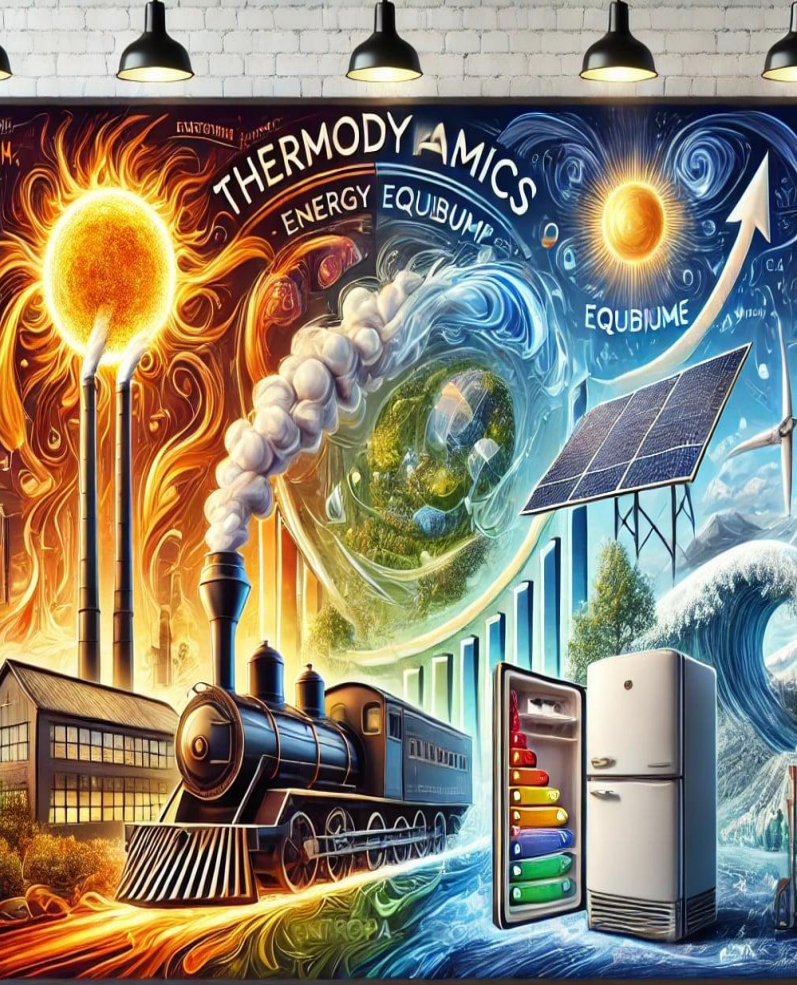




Laws of Thermodynamics



Thermodynamics:

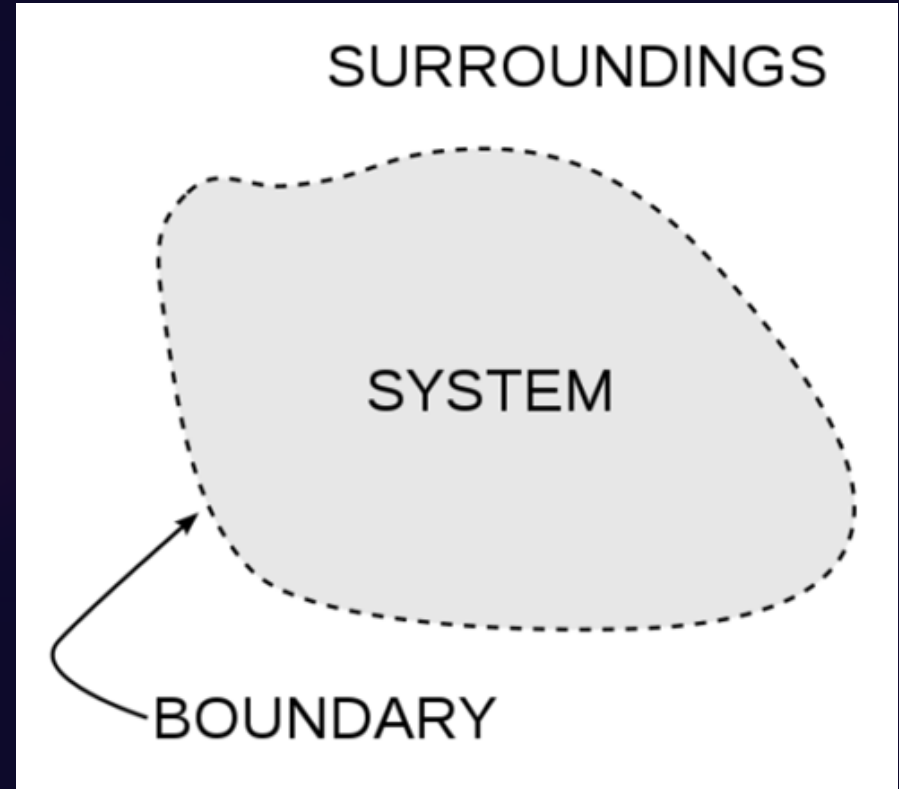
Thermodynamics is the study of the relations between heat, work, temperature, and energy.

Laws of Thermodynamics

- 1 Zeroth Law
- 2 First Law
- 3 Second Law
- 4 Third Law



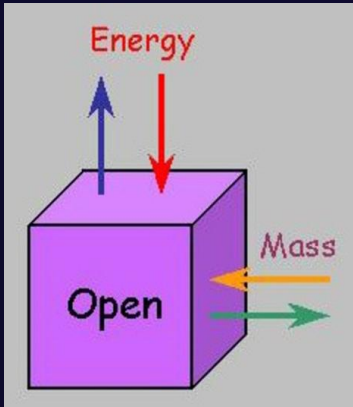
- 1 System
- 2 Surrounding
- 3 Boundary



Types of Thermodynamic Systems

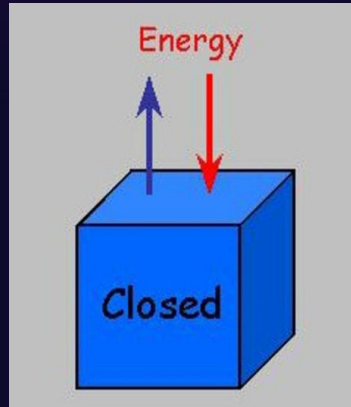
Open System

Can exchange both energy and mass with its surroundings. For example, a boiling pot of water.



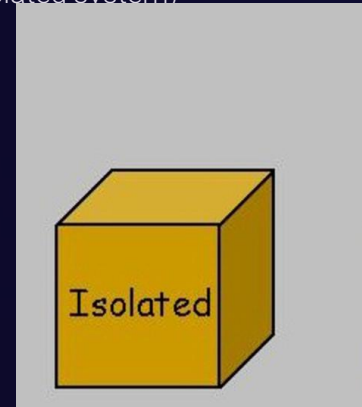
Closed System

Can exchange energy with its surroundings but not mass. For example, a sealed container of gas.



Isolated System

Cannot exchange either energy or mass with its surroundings. This is a theoretical construct. A thermos flask is an example. (not a perfect isolated system)



Internal Energy

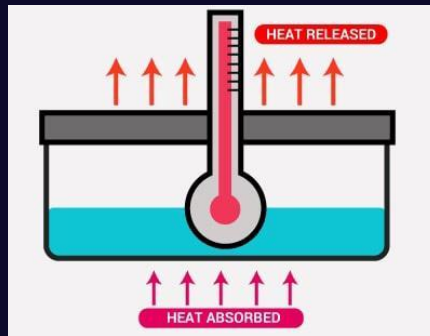
- ↪ It is donated by E or U
- ↪ depends on temperature and pressure
- ↪ Change in internal energy:

$$\Delta E = E(\text{production}) - E(\text{reaction})$$
$$= E_p - E_r$$



Enthalpy

- ↪ Heat content
- ↪ Change in enthalpy:
 $\Delta H = H(\text{products}) - H(\text{reactants})$
 $= H_p - H_r$



Entropy

- ↪ Disorder
- ↪ Melting of ice

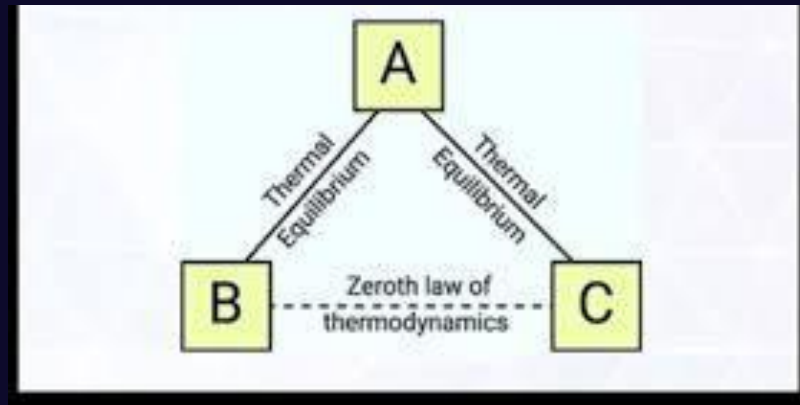


Zeroth Law of Thermodynamics

➡ ***System A + System B = System C***

➡ ***Transitive***

➡ ***Empirical parameter***



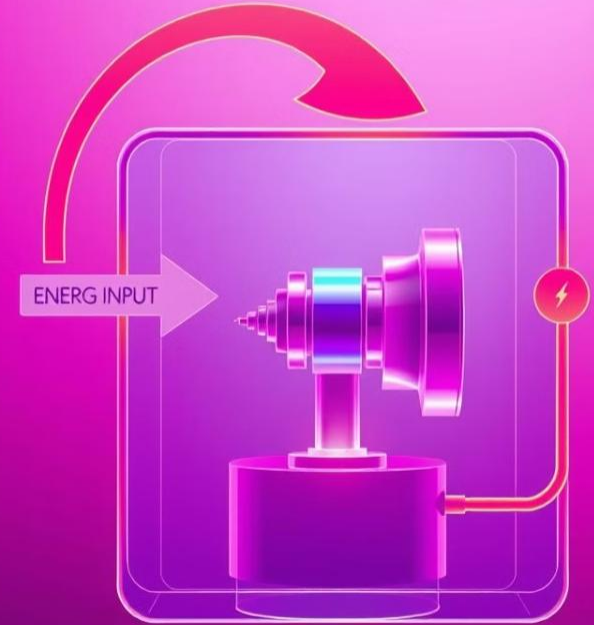
First Law of thermodynamics

Definition:

The First Law of Thermodynamics is a statement of the law of energy conservation:

"Energy cannot be created or destroyed; it can only be transferred or converted from one form to another."

This law applies to thermodynamic systems and explains how heat, work, and internal energy interact.





WORK

Mathematical Formulation:

$$\Delta U = Q - W$$

Where:

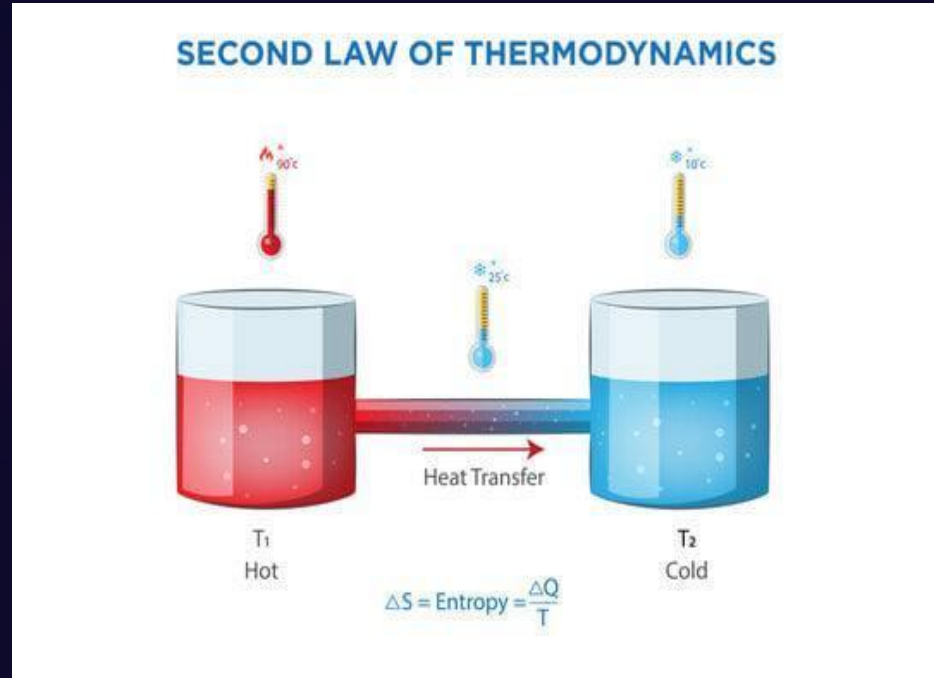
ΔU = Change in internal energy of the system

Q = Heat added to the system (+ve if heat is added, -ve if heat is removed)

W = Work done by the system (+ve if work is done by the system, -ve if work is done on the system)

Second Law of Thermodynamics

Heat can spontaneously flow from higher Temperature region to lower temperature region. But not the other way around





Entropy :

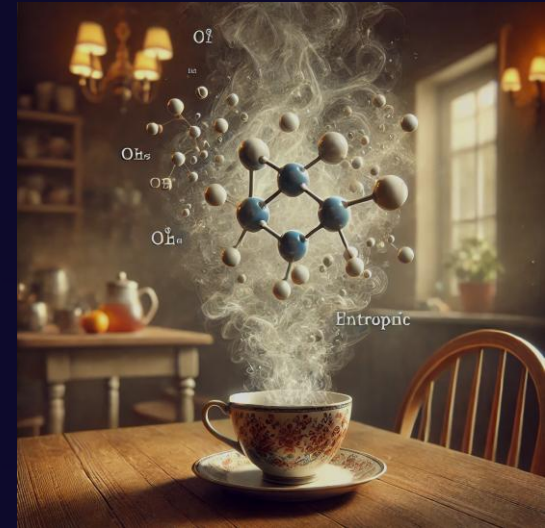
In any spontaneous process, the entropy of the universe always increases.

Irreversible process:

Entropy, $\Delta S > 0$

Reversible process :

Entropy, $\Delta S = 0$



Third Law of thermodynamics

The entropy of a system approaches a constant value as the temperature approaches absolute zero.

At zero temperature, the system must be in a state with the minimum thermal energy. This statement holds true if the perfect crystal has only one state with minimum energy. Entropy is related to the number of possible microstates according to:

$$S = k_B \ln \Omega$$

Where S is the entropy of the system, k_B is Boltzmann's constant, and Ω is the number of microstates.



Thank You !